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Alternative indexing: market cap or monkey?



Which index?



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- For many years investors have benchmarked their equity fund managers using market capitalisation-weighted indices
- Other, passive investors have chosen to track these indices
- A market capitalisation-weighted index gives the biggest weight to the constituent with the largest market capitalisation
- However, there are now a number of possible alternatives to this approach

“An evaluation of alternative equity indices, Part 1: Heuristic and optimised weighting schemes”, and “An evaluation of alternative equity indices, Part 2: Fundamental weighting schemes”, by Clare, A., N. Motson, & S. Thomas, Cass Business School, March 2013.

This research is based upon two papers commissioned by Aon Consulting. The papers can be downloaded from:
<http://ssrn.com/abstract=2242034> & <http://ssrn.com/abstract=2242028>

The set of alternatives I



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- “Heuristic” approaches:
 - Equally-weighted
 - Diversity-weighted
 - Inverse volatility
 - Equal risk contribution
 - Risk clustering

Diversity weights example



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- Diversity weighting is a half way house between a cap weighting and an equal weighting scheme
- The market cap weight of each constituent, is raised to the power “p”
- W_i^P if P is set to 1 then the weight is just the market cap weight; if P is set to 0 then every constituent has the same weight (ie, an equal weight)

An example of Diversity Weighting for an index with five stocks

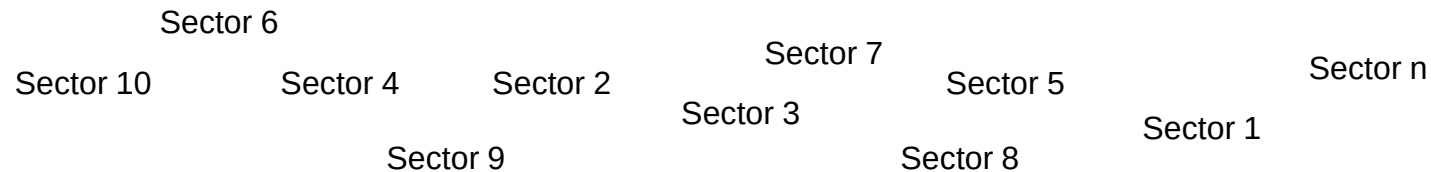
	Market cap	MCW (1)	DW (0.75)	DW (0.50)	DW (0.25)	EW (0)
Stock A	100	54.1%	44.9%	35.8%	27.3%	20%
Stock B	35	18.9%	20.4%	21.2%	21.0%	20%
Stock C	15	8.1%	10.8%	13.9%	17.0%	20%
Stock D	10	5.4%	8.0%	11.3%	15.4%	20%
Stock E	25	13.5%	15.9%	17.9%	19.3%	20%
	185	100%	100%	100%	100%	100%

Risk clustering example

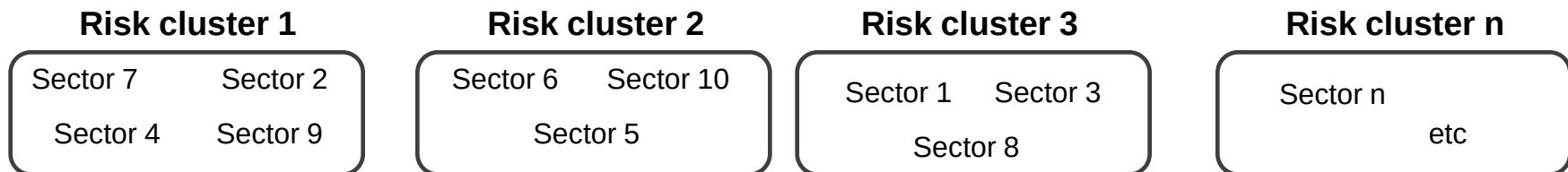


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- Identify your market-cap weighted sectors ...



- Place them in equally-weighted risk clusters



- Then assign each risk cluster an equal weight. Probably works best at an international level
- A simpler version of this might be to equally weight industrial sectors, but where stocks are market cap weighted within each sector

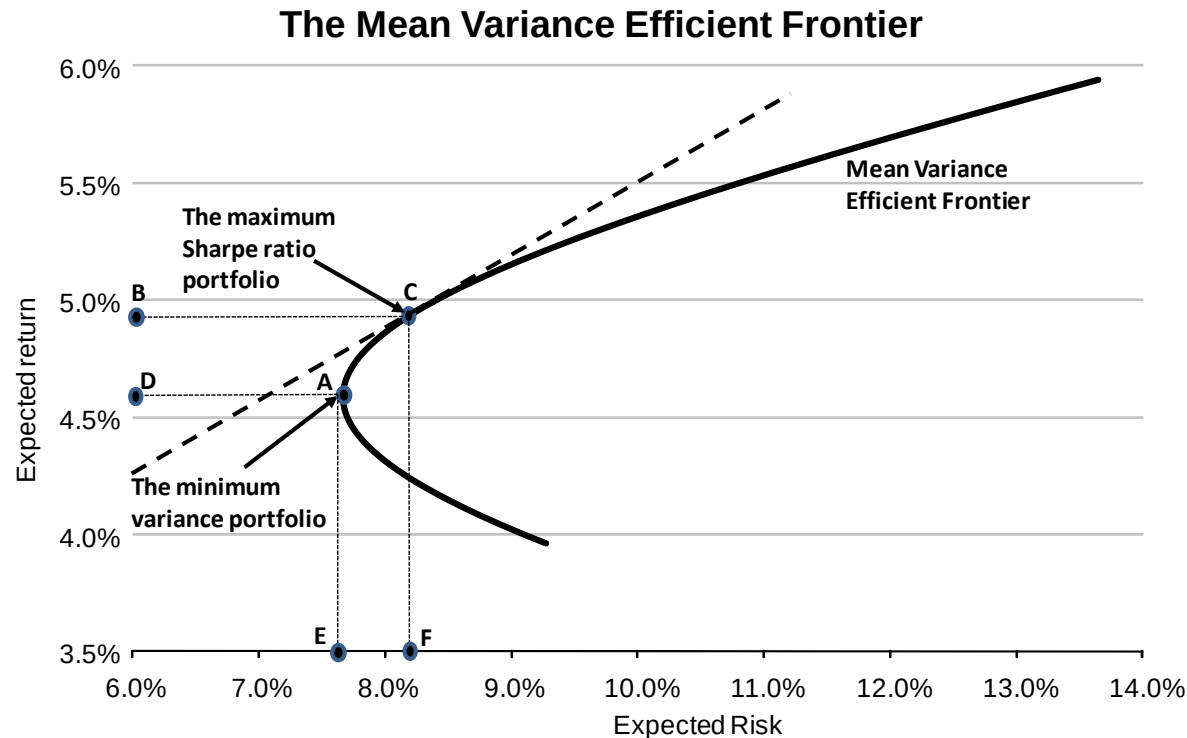
The set of alternatives II

- Optimised approaches to index construction
- These are more complex and require maximisation, or minimisation of a mathematical function
 - **Minimum Variance weights**
 - **Maximum Diversification weights**
 - **Risk Efficient weights**
- Constraints are set so that the optimisation process does not come up with extremely concentrated portfolios – such as the maximum amount to be invested in any one constituent

Optimised weights



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- The minimum variance approach identifies the weights of the stocks that comprise portfolio A above – the minimum variance portfolio
- This process might suit those that believe that the expected return on every constituent is identical

Maximum Diversification weights



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- But why hold the lowest risk, lowest expected return efficient portfolio?
- The *Maximum Diversification* Approach seeks to identify the weights that produce portfolio C on the efficient frontier
- But to do this they need to calculate the expected return on each constituent. How do they do this?
- They assume (a heuristic assumption) that expected return is linearly related to stock volatility – *the more volatile a stock the higher its expected return*
- They then maximise the following expression:

$$\frac{\text{Weighted average constituent standard deviation}}{\text{Standard deviation of portfolio}}$$

Risk Efficient weights



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- Risk Efficient weights are determined in a similar manner. But the expected return on each constituent is assumed (another heuristic assumption) to be linearly related to the semi-deviation of its return; they:
 - calculate the semi-deviation of the return on each stock
 - group them in to deciles, and calculate the average semi-deviation of each decile
 - then every constituent in, for example, decile 1, is assigned the “expected return” of its decile, etc
- They then maximise the following function to find their version of portfolio C:

$$\frac{\text{Weighted average constituent semi - deviation}}{\text{Standard deviation of portfolio}}$$

The set of alternatives III



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- “Fundamentally-weighted” approaches:
 - Dividend-weighted
 - Cashflow-weighted
 - Book value-weighted
 - Sales-weighted
 - Composite
- These are just alternative measures of company scale

Data and methodology I

- Every year, from 1968 to 2011, we gathered data from the CRSP data files on the largest 1,000 US stocks that had five years of continuous total return history
- At the end of the first year (1968) we applied the weights according to the rules of each index construction methodology
- We then calculated the returns and related information on each index over 1969
- We repeated this process at the end of each year, until we had constructed a continuous time series of the indices from Jan 1969 to Dec 2011
- The indices were all therefore rebalanced annually



- For the heuristic and optimised indices that required the calculation of historic volatilities we used 5 years of historic data to calculate the relevant terms
- For the Diversity Weighting index we set $P=0.76$ (we also tested other values of P)
- For the optimised indices we imposed a constituent weight cap of 5% (we also tested other caps and restrictions on constituent weights)

Data and methodology III



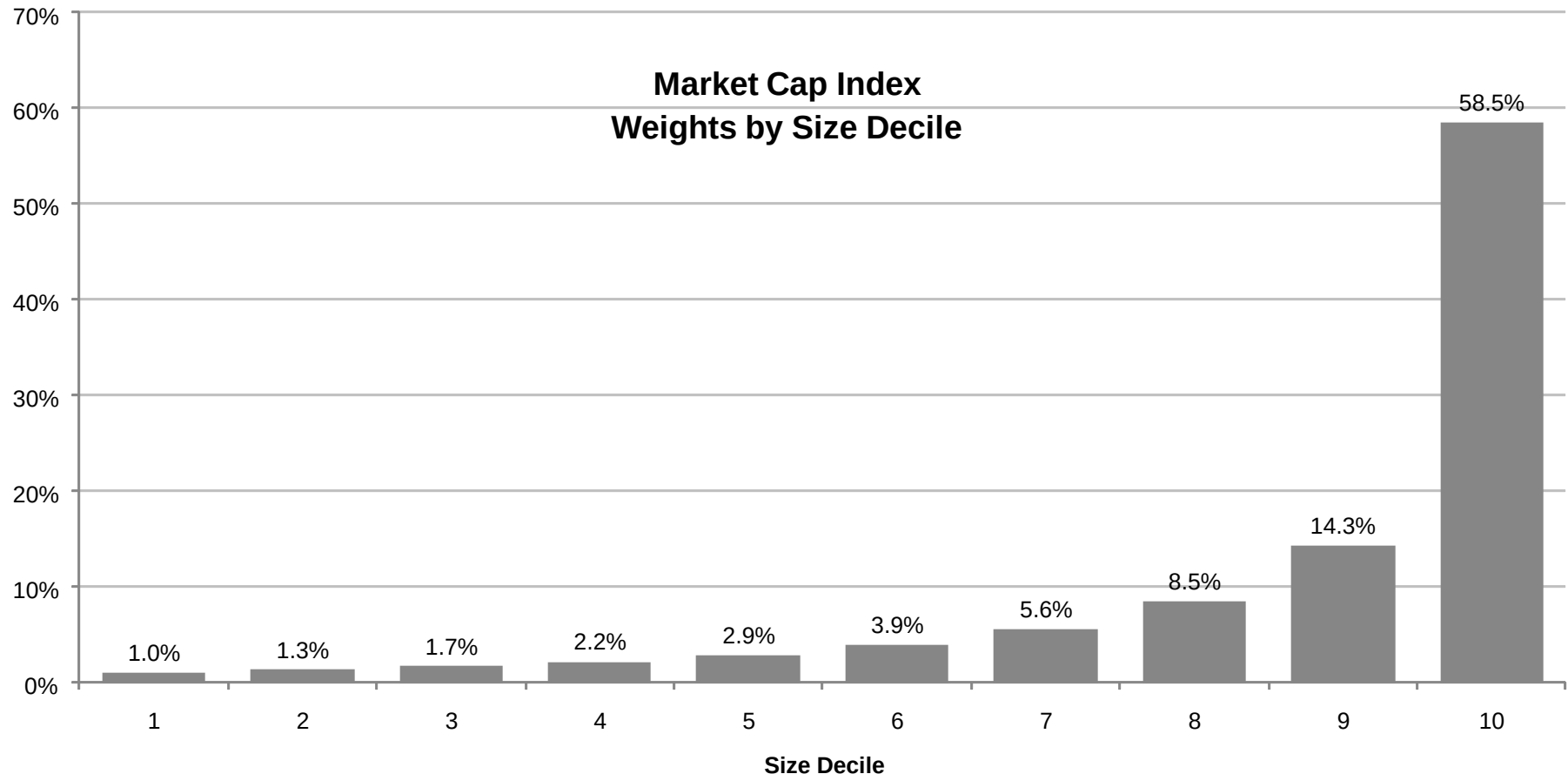
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- For the Dividend-weighted index we summed the total dividend for each stock over the previous five years. The weight for each constituent was this sum, divided by the sum of this value for all 1,000 stocks
- We applied the same approach to calculate the Sales, Book-value and Cashflow weights
- We also constructed a Composite index, where we calculated the average Dividend weight, Sales, Book-value and Cashflow weight that each stock had and used this as the composite index weight

How concentrated is the market cap index?



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- The largest 100 stocks make up over half the index and the largest 200 make up almost three quarters

Full sample results



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	Return	Standard Deviation	Sharpe Ratio
Market cap weighted	9.4%	15.3%	0.32
Equal -Weighted	11.0%	17.2%	0.39
Diversity Weighting	10.0%	15.7%	0.35
Inverse Volatility	11.4%	14.6%	0.45
Equal Risk Contribution	11.3%	15.6%	0.43
Risk Clustering	9.8%	16.7%	0.33
Minimum Variance	10.8%	11.2%	0.50
Maximum Diversification	10.4%	13.9%	0.40
Risk Efficient	11.5%	16.7%	0.42
Dividend -weighted	10.8%	14.5%	0.42
Cashflow -weighted	10.9%	15.2%	0.41
Book Value -weighted	10.7%	15.7%	0.39
Sales -weighted	11.4%	16.2%	0.42
Fundamentals Composite	11.0%	15.3%	0.41

- All 13 of the alternative indices have a higher return; 6 out of 13 have lower volatility; and all 13 have a higher Sharpe Ratio

The 1970s and 1980s



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	1970s			1980s		
	Return	Standard Deviation	Sharpe Ratio	Return	Standard Deviation	Sharpe Ratio
Market cap weighted	6.1%	16.2%	0.07	16.9%	16.1%	0.53
Equal-Weighted	9.0%	19.9%	0.22	17.8%	16.7%	0.56
Diversity Weighting	6.9%	17.1%	0.12	17.1%	16.2%	0.54
Inverse Volatility	9.4%	17.1%	0.25	19.6%	14.6%	0.72
Equal Risk Contribution	9.3%	18.4%	0.24	18.9%	15.5%	0.65
Risk Clustering	6.4%	18.4%	0.10	17.8%	17.3%	0.55
Minimum Variance	7.8%	12.9%	0.17	20.2%	12.0%	0.89
Maximum Diversification	7.5%	16.8%	0.15	20.0%	13.6%	0.79
Risk Efficient	9.6%	20.0%	0.25	18.6%	16.1%	0.62
Dividend-weighted	8.7%	15.4%	0.22	19.1%	14.3%	0.71
Cashflow-weighted	9.2%	16.1%	0.25	18.6%	15.4%	0.64
Book Value-weighted	9.1%	16.4%	0.24	18.3%	15.4%	0.62
Sales-weighted	9.1%	17.6%	0.23	19.4%	16.2%	0.66
Fundamentals Composite	9.0%	16.3%	0.23	18.8%	15.3%	0.66

- The cap-weighted index underperforms across both decades

The 1990s and Noughties



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	1990s			2000s		
	Return	Standard Deviation	Sharpe Ratio	Return	Standard Deviation	Sharpe Ratio
Market cap weighted	17.6%	13.1%	0.94	0.4%	15.2%	-0.07
Equal-Weighted	15.0%	13.7%	0.74	6.2%	17.0%	0.28
Diversity Weighting	17.1%	13.1%	0.91	2.6%	15.5%	0.07
Inverse Volatility	13.2%	11.6%	0.72	6.9%	14.2%	0.35
Equal Risk Contribution	14.0%	12.5%	0.73	6.6%	15.1%	0.32
Risk Clustering	13.5%	13.3%	0.66	5.1%	16.5%	0.22
Minimum Variance	11.2%	9.8%	0.65	6.5%	10.4%	0.39
Maximum Diversification	12.7%	11.7%	0.67	4.6%	12.4%	0.21
Risk Efficient	14.9%	13.5%	0.74	7.0%	16.1%	0.33
Dividend-weighted	15.4%	11.7%	0.88	4.0%	15.5%	0.15
Cashflow-weighted	16.4%	12.0%	0.93	3.2%	16.4%	0.11
Book Value-weighted	17.0%	12.9%	0.91	2.9%	17.0%	0.10
Sales-weighted	17.0%	13.2%	0.90	4.2%	16.9%	0.16
Fundamentals Composite	16.5%	12.4%	0.91	3.6%	16.3%	0.13

- Cap-weighted index is the star in the 1990s but performs badly in Noughties

How smart is “smart beta” ?



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	Alpha	Beta
Market cap weighted	0.0000	1.00
Equal-Weighted	0.0009	1.06
Diversity Weighting	0.0004	1.02
Inverse Volatility	0.0024*	0.89
Equal Risk Contribution	0.0018*	0.96
Risk Clustering	0.0002	1.03
Minimum Variance	0.0048*	0.51
Maximum Diversification	0.0020*	0.82
Risk Efficient	0.0020*	0.82
Dividend-weighted	0.0019*	0.89
Cashflow-weighted	0.0015*	0.96
Book Value-weighted	0.0011*	0.99
Sales-weighted	0.0015*	1.02
Fundamentals Composite	0.0015*	0.97

- Pretty smart !!!

How smart is “smart beta” ?



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	Alpha	Beta Rm-Rf	Beta SMB	Beta HML
Market cap weighted	0.0001	0.9748*	-0.1212*	0.0517*
Equal-Weighted	-0.0001	1.0131*	0.2868*	0.2887*
Diversity Weighting	0.0002	0.9897*	-0.0264*	0.1243*
Inverse Volatility	0.0003	0.8916*	0.0989*	0.3913*
Equal Risk Contribution	0.0001	0.9468	0.1784*	0.3465*
Risk Clustering	-0.0005	1.0103*	0.0662*	0.2456*
Minimum Variance	0.0008	0.5611*	-0.0292	0.4316*
Maximum Diversification	0.0002	0.7981*	0.1673*	0.2510*
Risk Efficient	0.0010	0.8394*	0.2404*	0.4197*
Dividend-weighted	-0.0001	0.9383*	-0.1569*	0.4169*
Cashflow-weighted	0.0001	0.9852*	-0.1053*	0.3360*
Book Value-weighted	-0.0002	1.0139*	-0.0798*	0.3573*
Sales-weighted	0.0001	1.0299*	-0.0046	0.3775*
Fundamentals Composite	0.0000	0.9919*	-0.0866*	0.3719*

- Maybe not so smart !!!



"If one puts an infinite number of monkeys in front of (strongly built) typewriters and lets them clap away (without destroying the machinery), there is a certainty that one of them will come out with an exact version of the 'Iliad.' Once that hero among monkeys is found, would any reader invest [their] life's savings on a bet that the monkey would write the 'Odyssey' next?" (N. Taleb)

A simian experiment



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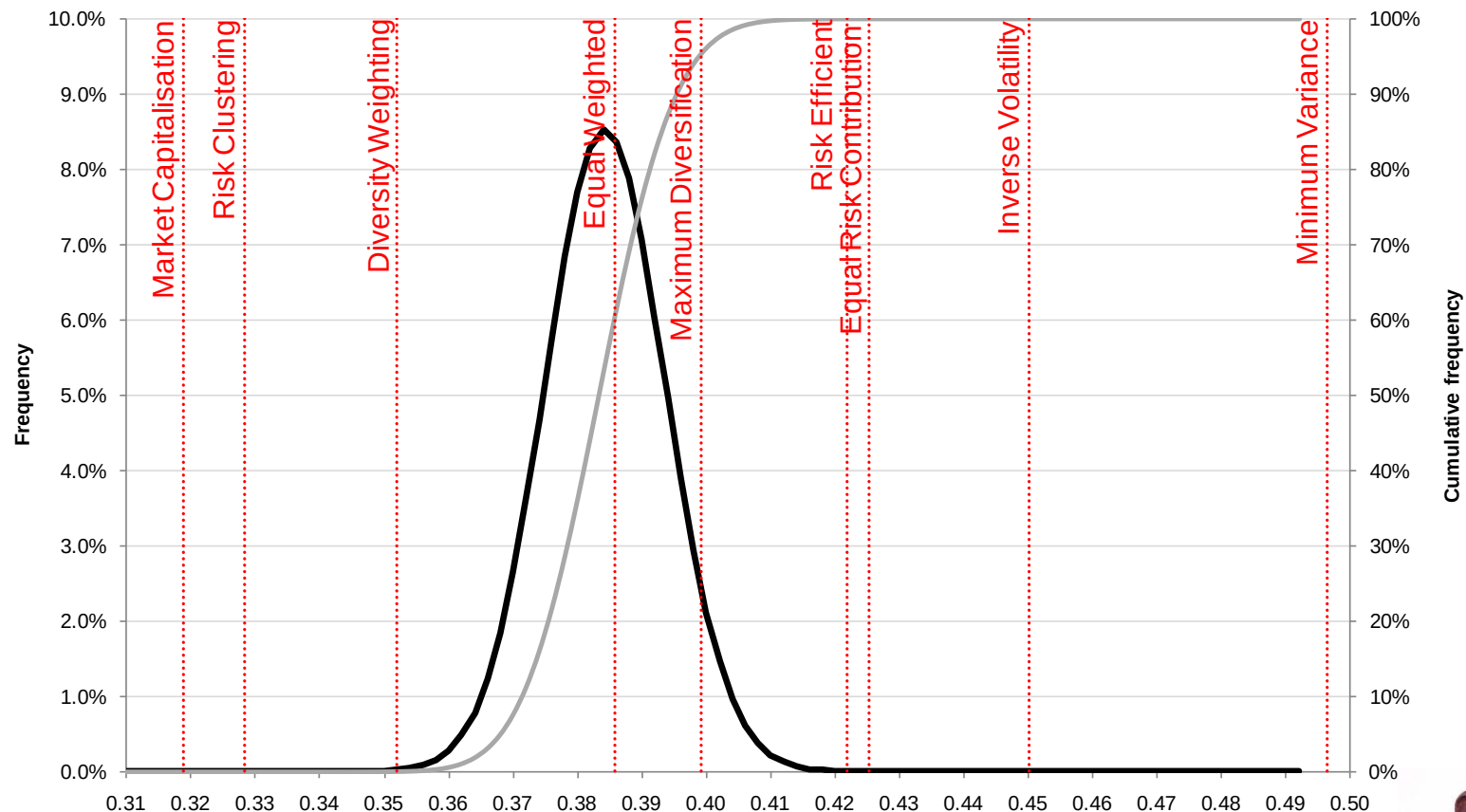
- Are the alternative indices good, or did they just get lucky?
- We designed a simple experiment, at the end of each year we 'asked' the computer to:
 - choose, at random one stock of the 1,000
 - we assigned this stock a weight of 0.1%
 - we did this 1,000 times: if a stock was chosen once it was assigned a weight of 0.1%, if not at all, then a weight of 0.0%, if 1,000 times, then a weight of 100%
 - we then repeated this for every year in our sample, which ultimately produced an index that may just as well have been chosen by a monkey
 - We then repeated this entire process 10,000,000 times !!!
 - Giving us 10,000,000 randomly chosen indices



I. A simian experiment: Sharpe ratio



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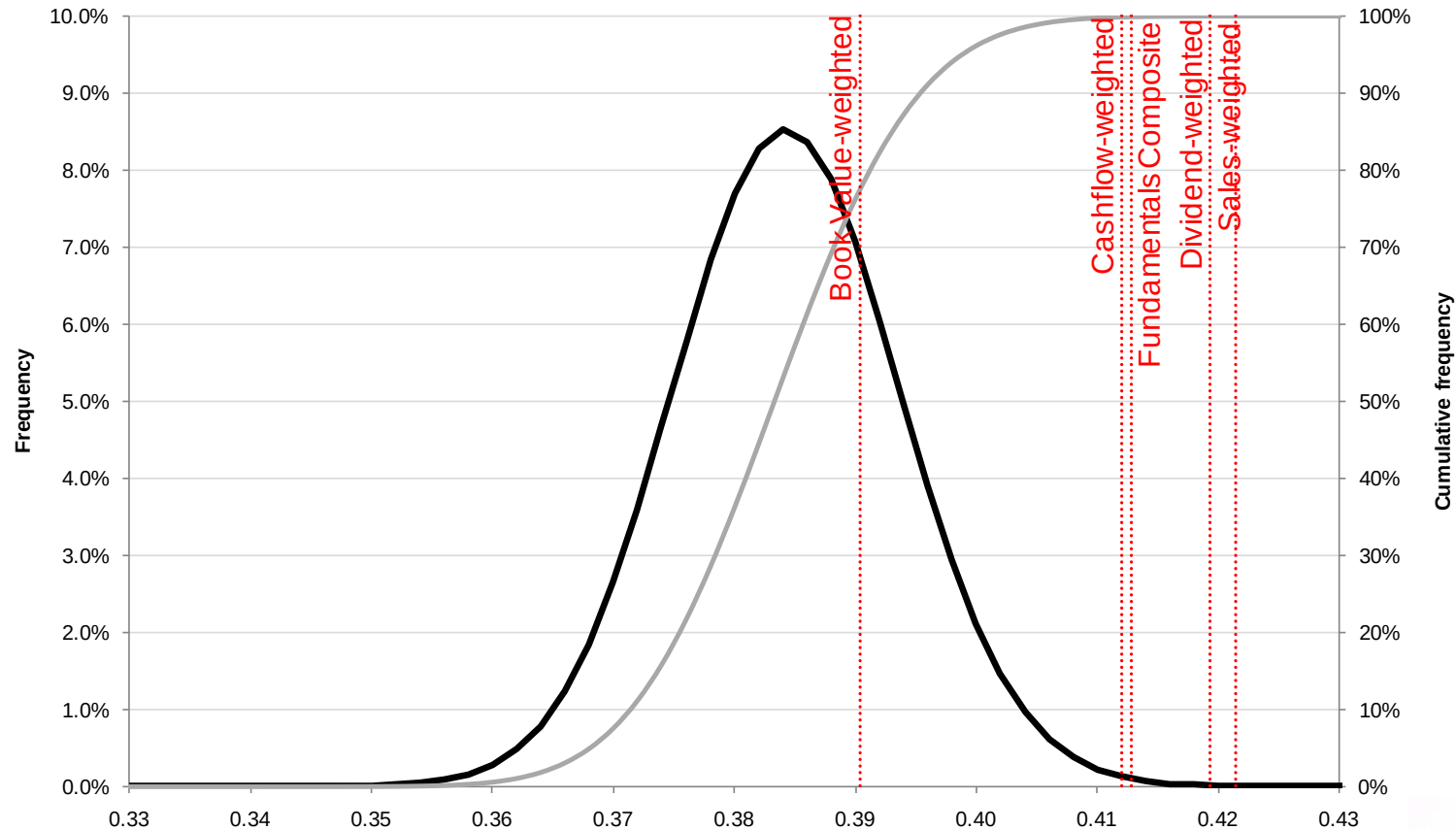
- The Sharpe ratio of the MVP-weighted index is really off the scale



II. A simian experiment: Sharpe ratio



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- The sales-weighted Sharpe is better than most of those produced by the 10m monkeys



Is there any hope for cap-weighted indices?



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- A cap-weighted index:
 - is comprised of more liquid stocks and involves much less turnover
 - seems to outperform in a bull market
 - And can be much improved with the addition of a simple risk filter ...



Applying TF to a market cap-weighted index



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- Aon Consulting also asked us to explore the possibility of incorporating market timing rules into index construction
- We picked the simplest one, that has been established in the academic literature as being potentially useful:
- The rule that we applied was very simple:
 - at the end of the month if the index value was greater than its ten month moving average, we 'invested' 100% in the equity index and earned the return on that index in the following month;
 - but if at the end of the month the index value was lower than its ten month moving average, we 'invested' 100% in US T-bills and earned the T-bill return in the following month.

* The Trend is Our Friend: Risk Parity, Momentum and Trend Following in Global Asset Allocation, A. Clare, J. Seaton, P. Smith and S. Thomas, <http://ssrn.com/abstract=2126478>



Timing is everything (full sample) ...



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Panel A: Full sample results (1969 to 2011)								
	Return	St. dev.	Sharpe Ratio	Sortino Ratio	Max Draw down	% Positive Months	Alpha	Beta
Market cap weighted	10.5%	11.6%	0.46	0.55	-23.3%	73.8%	0.7%	0.53
Equal-weighted (2.1)	10.3%	12.6%	0.41	0.50	-27.7%	71.7%	0.6%	0.55
Diversity weighting (2.2)	10.4%	11.9%	0.44	0.53	-23.4%	72.8%	0.6%	0.55
Inverse Volatility (2.3)	10.4%	11.1%	0.46	0.54	-21.8%	72.6%	0.7%	0.49
Equal risk contribution (2.4)	10.0%	11.9%	0.41	0.48	-23.2%	72.6%	0.6%	0.53
Risk clustering (2.5)	8.8%	12.2%	0.31	0.37	-25.7%	71.3%	0.5%	0.51
MVP-weighted (3.1)	11.6%	8.7%	0.69	0.80	-16.8%	76.2%	0.8%	0.28
Maximum diversification weights (3.2)	9.5%	10.6%	0.40	0.45	-20.2%	72.0%	0.6%	0.46
Risk Efficient (3.3)	11.0%	11.7%	0.49	0.61	-25.7%	73.8%	0.7%	0.45

Panel A: Full sample results (1969 to 2011)								
	Returns	St. dev.	Sharpe Ratio	Sortino Ratio	Max Draw down	% Positive Months	Alpha	Beta
Market cap-weighted	10.5%	11.6%	47.0%	56.3%	-23.3%	73.8%	0.4%	56.7%
Dividend-weighted	11.1%	10.8%	54.6%	66.0%	-20.7%	72.4%	0.5%	51.0%
Cashflow-weighted	11.1%	11.4%	52.0%	62.3%	-22.7%	72.8%	0.5%	54.8%
Book Value-weighted	10.7%	11.7%	47.9%	57.9%	-23.0%	72.6%	0.4%	56.1%
Sales-weighted	10.9%	12.1%	48.2%	58.1%	-24.5%	72.0%	0.4%	57.2%
Fundamentals composite-weighted	11.1%	11.4%	52.1%	62.4%	-22.7%	72.8%	0.5%	54.8%

- The rule improves Sharpe ratios and reduces maximum drawdowns by around half



Timing is everything (by decade) ...



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Panel B: Annualised returns and volatility by decade								
	1970s		1980s		1990s		2000s	
	Return	St. dev.	Return	St. dev.	Return	St. dev.	Return	St. dev.
Market cap weighted	9.4%	11.1%	14.6%	14.0%	14.7%	12.1%	7.5%	8.3%
Equal-weighted (2.1)	8.1%	13.9%	13.9%	14.2%	13.6%	11.3%	9.1%	10.6%
Diversity weighting (2.2)	8.6%	11.7%	15.3%	14.2%	14.2%	12.0%	7.5%	8.9%
Inverse Volatility (2.3)	9.8%	12.1%	13.5%	12.8%	11.9%	10.1%	9.2%	9.5%
Equal risk contribution (2.4)	8.6%	12.8%	13.9%	13.4%	11.0%	11.3%	9.6%	9.8%
Risk clustering (2.5)	7.6%	12.6%	12.7%	14.8%	9.8%	10.9%	8.5%	10.0%
MVP-weighted (3.1)	7.7%	8.1%	19.1%	10.8%	13.3%	8.1%	6.9%	7.4%
Maximum diversification weights (3.2)	8.8%	11.3%	14.5%	12.4%	10.4%	10.3%	6.8%	7.7%
Risk Efficient (3.3)	9.8%	12.9%	14.6%	13.6%	10.1%	10.7%	10.5%	9.7%

Panel B: Annualised Returns and volatility by decade								
	1970s		1980s		1990s		2000s	
	Return	St. dev.	Return	St. dev.	Return	St. dev.	Return	St. dev.
Dividend-weighted	8.7%	11.5%	15.9%	12.8%	12.0%	10.2%	10.4%	9.0%
Cashflow-weighted	9.8%	12.4%	16.6%	13.5%	11.9%	10.3%	8.8%	9.6%
Book Value-weighted	9.7%	12.5%	15.8%	13.8%	11.7%	11.1%	8.0%	9.9%
Sales-weighted	7.8%	13.2%	16.3%	14.5%	11.8%	11.3%	9.8%	10.0%
Fundamentals composite-weighted	9.6%	12.5%	16.3%	13.6%	11.4%	10.6%	9.5%	9.5%

- The rule has a particularly significant impact on the cap-weighted index in the Noughties, but improves the performance of all indices over this period



Summary



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- The performance of cap-weighted indices has been disappointing particularly over the Noughties – even a monkey could have done better!
- Transactions costs cannot explain the performance difference
- Many of the alternatives offer the possibility of far superior risk-adjusted performance
- So these indices all offer passive investors an alternative to the cap-weighted indices that they have been tracking for many years now ...



Simian Asset
Management

“Go on index trackers,
make my day!”



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